

Energy Law — Module 1: Comprehensive Study Document

PART A — Foundations of Energy Law

1.1 What Is Energy Law?

Core definition (PPT): The legal rules, institutions, and processes governing the entire energy life-cycle — exploration and extraction, conversion, transport, distribution, consumption, decommissioning, and waste.

Bradbrook (1996) gives the classic definition: energy law concerns “the allocation of rights and duties concerning the exploitation of all energy resources between individuals, between individuals and the government, between governments and between States.” This definition is deliberately broad: it covers both **domestic** and **international** dimensions, and both **private** and **public** law. The four relationships Bradbrook identifies are:

1. **Private–private:** e.g., a power purchase agreement (PPA) between a generator and a distribution company.
2. **Private–State:** e.g., a mining licence or a production sharing contract (PSC).
3. **Union–State** (in federal systems like India): the constitutional division of competence under the Concurrent List.
4. **State–State:** treaties on transit, investment protection, and climate change.

Why this definition matters for the exam: It establishes that energy law is not merely “electricity law” or “oil and gas law.” It is a **cross-cutting field** that spans constitutional law, administrative law, contract law, international law, and environmental law. The life-cycle framing is critical: energy law does not stop at the meter — it extends to decommissioning and waste (a point Heffron and Talus later emphasise as the “fifth stage” of energy law’s evolution).

1.2 The Dual Meaning of Energy Law (Huhta)

Energy law has two distinct meanings that must not be conflated:

Meaning	What it refers to	Age and character
Normative system	The body of substantive and procedural norms that govern the sector: constitutions, treaties, statutes, regulations, contracts, judgments	Over a century old (Indian Electricity Act 1910)
Legal discipline	The academic field that studies, systematises, critiques and teaches those norms	Only two to three decades old

Why the distinction matters: The normative system predates the academic discipline by decades. This is not unique to energy law — criminal law and property law existed long before they became university subjects — but it has a specific consequence for energy law: scholars in the 1990s and 2000s were often describing and justifying rules that regulators and courts had already made. The discipline is thus **reactive as much as prescriptive**.

The two meanings **shape each other**:

- **Norms → research:** The unbundling and third-party access rules of the 1990s became central research themes.

- **Research → norms:** Scholarship is cited in regulators' consultation papers and in judgments. *Energy Watchdog v. CERC* (2017) 14 SCC 80 on force majeure and change in law is the leading Indian example.

Heffron and Talus (2016) argue that for energy law to fully emerge as its own sub-discipline, it needs its own **theory** – not just descriptive accounts of statutes and cases. They propose three theoretical frameworks:

1. **The Energy Law and Policy Triangle** (a precursor to the trilemma framework).
2. **The Theory of Change in Energy Law** (how legal rules evolve in response to external shocks).
3. **The Power of Energy Law: Targeted Legislation** (how law can be designed to achieve specific energy-policy outcomes).

This is a key point for exam essays: energy law is still **in the process of becoming** a coherent discipline. It borrows from contract, administrative, competition, environmental, and investment law, and has not yet developed the kind of internal coherence that criminal law or tort law possesses.

1.3 Five Characteristics of the Discipline

The PPT identifies five features that distinguish energy law as a field:

Characteristic	Meaning	Consequence
Problem-based	Starts from concrete disputes, not abstract theory	Reach beyond the “energy statute”
Fragmented, multilevel governance	Competence spread across international, Union, State and local bodies	Map who decides what
Borrowing	Absorbs contract, administrative, competition, environmental, investment law	Not autonomous like criminal law
Plural, conflicting objectives	Every decision balances security, cost, sustainability, sovereignty	The trilemma (Part B)
Interdisciplinary	Questions are also physics, engineering, economics	Respect the physical system

The governance fragmentation point is worth expanding. In India, a single energy project touches: - **International:** the Paris Agreement and India's NDC. - **Union:** MNRE guidelines, CERC inter-State transmission rules. - **State:** State ERC tariff and open-access regulations. - **Local:** district land allotment and gram panchayat consent. - **Public corporation:** SECI as PPA counterparty. - **Private:** lenders' covenants.

A dispute at any layer can stall the project. This is why energy law scholarship often focuses on **jurisdictional conflicts** as much as on substantive rules.

The M.K. Ranjitsinh v. Union of India litigation is the perfect illustration of interdisciplinarity. In April 2021, the Supreme Court ordered power lines in Great Indian Bustard habitat in Rajasthan and Gujarat to go underground. Developers argued that undergrounding high-voltage lines was technically infeasible. In April 2024, the Court modified the blanket direction, set up an expert committee, and recognised a **right against the adverse effects of climate change** under Articles 14 and 21. Deciding the case required engineering, ecology, and climate science – not just legal doctrine.

1.4 Evolution: Five Stages (Heffron & Talus 2016)

The PPT presents a five-stage evolutionary model. This model is the central analytical contribution of Heffron and Talus to the field.

Stage	Driver	Illustrative Legislation
1. Safety (1820s→)	Mining accidents; worker protection	UK Mines and Collieries Act 1842; Indian Mines Act 1901; Mines Act 1952
2. Security (1910s→)	Preventing war: control of coal and steel	Treaty of Paris 1951 (ECSC); Euratom Treaty 1957
3. Economics (1960s→)	1970s oil shocks; liberalisation ideology	UK Electricity Act 1989; EU Directive 96/92/EC; India's Electricity Act 2003
4. Infrastructure (2000s→)	Under-investment revealed by markets	EU Projects of Common Interest; India's Green Energy Corridor
5. Justice (emerging)	Fairness in process and outcome; the waste phase	Paris Agreement preamble; EU Just Transition Fund 2021; Jharkhand Just Transition Task Force 2022

Critical qualification: These stages **overlap and coexist**. India in 2026 simultaneously operates under Stage 2 (strategic petroleum reserves), Stage 3 (competitive bidding for renewables), Stage 4 (Green Energy Corridor), and Stage 5 (just transition planning in coal districts). The model is not a strict chronology but a **layering** of concerns.

Why the fifth stage matters: Heffron and Talus argue that energy law has historically neglected the **waste management** phase of the energy life-cycle. Decommissioning of nuclear plants, disposal of solar panels and batteries, and rehabilitation of coal mines are only now becoming central legal concerns. The “justice” stage is thus both about **distributional fairness** (who bears the costs of transition) and about **life-cycle completeness** (the waste phase must be legally governed, not an afterthought).

1.5 Energy Law and Its Neighbours

The PPT distinguishes energy law from three adjacent fields:

Feature	Energy Law	Energy Policy	Energy Economics	Environmental Law
Focus	Binding rights, duties, procedures	Goals and strategic direction	Prices, markets, incentives	Protection of the environment
Key question	Is it permissible and enforceable?	What should we achieve?	What will actors do?	What are the ecological limits?
Method	Doctrinal: statute, contract, precedent	Political science, administration	Econometrics, cost-benefit	Standards, impact assessment, tort
Indian output	Acts, regulations, licences, judgments	National Electricity Policy 2005; NDC	Tariff design, forecasts	EIA Notification 2006; NGT orders

The crucial distinction: Policy sets objectives; law converts them into **enforceable obligations**. For example: - **Policy:** 500 GW non-fossil capacity by 2030. - **Law:** Renewable Purchase Obligation (RPO) under s. 86(1)(e), Electricity Act 2003.

Economics explains whether the obligation is efficient. Environmental law constrains the sector from outside without concern for supply as such. Legal rules are **coercive**: a thermal plant without environmental clearance faces stop-work orders, penalties under the EPA 1986, and NGT directions — not merely a recommendation.

The RPO as a four-lens illustration: - **Policy:** National Action Plan on Climate Change (2008) and successive NDCs set a target share of renewables. - **Law:** s. 86(1)(e) requires State Commissions to fix a minimum renewable share for DISCOMs and open-access consumers. In *Hindustan Zinc v. Rajasthan ERC* (2015) 12 SCC 611, the Supreme Court upheld RPO even for captive plants: a **binding duty, not an aspiration**. - **Economics:** RPO creates a market for renewable energy certificates and shifts the merit order. - **Environment:** Displaced coal generation reduces regulated emissions.

1.6 Energy Law versus Resources Law

This is one of the most important conceptual distinctions in the module. It maps directly onto the **supply-security vs. demand-security** divide.

Dimension	Energy Law and Policy	Resources Law and Policy
Core objective	Reliable supply at least cost	Maximum value and control for the owner
Whose concern	Consuming and importing States	Producing and exporting States
Security concept	Security of supply	Security of demand
Governance	Market-oriented: competition, regulators	State-oriented: ownership, licences, royalties
Instruments	Unbundling, third-party access, tariffs, stocks	PSCs, royalties, nationalisation, export controls

Two logics in practice:

- **Energy law: EU market liberalisation.** Directives from 1996 to the Clean Energy Package 2019; unbundling of networks from generation; regulated third-party access to grids; supplier choice; independent regulators (ACER). Goal: efficiency and security through competition.
- **Resources law: the production sharing contract.** Indonesia 1966: State keeps ownership of the oil; contractor bears risk and recovers “cost oil”; “profit oil” shared with the State. India: PSCs under NELP (1997–2016); revenue sharing under HELP (2016). The KG-D6 dispute shows how PSC accounting is contested.

Security of demand: OPEC versus the IEA. OPEC (1960) coordinates members’ output to influence prices. This is not about ensuring supply (members could produce more) but about keeping demand for their oil profitable – the resources-law concern with **security of demand**. The International Energy Agency (1974), created by consuming countries after the first oil shock with a 90-day stockholding obligation, embodies the energy-law concern with **security of supply**.

1.7 Sources of Energy Law (India)

The PPT presents a **hierarchy of sources**:

1. **International:** Treaties, custom, soft law, internationalised domestic models.
2. **Constitutional:** Seventh Schedule; Articles 21, 39(b), 48A, 254.
3. **Legislative:** Electricity Act 2003; Energy Conservation Act; MMDR; ORD; PNGRB; SHANTI Act 2025.
4. **Delegated and policy:** Grid Code; tariff regulations; National Electricity Policy; Tariff Policy; schemes.
5. **Judicial and administrative:** Supreme Court, High Courts, APTEL, NGT; ministry guidelines.

No single international body lays down a uniform energy code. This is a defining feature of international energy law: it is **fragmented** across treaties, custom, and soft law.

Key international treaties:

Treaty	Relevance	India
UNCLOS 1982	Sovereign rights over offshore oil, gas, wind in EEZ and continental shelf	Party (1995)
Energy Charter Treaty 1994	Investment protection, transit, trade; investor-State arbitration	Not a party; EU withdrew 2024
UNFCCC 1992 · Kyoto 1997 · Paris 2015	NDCs require decarbonising energy	Party; NDC 2022; new 2035 NDC (2026)
Convention on Nuclear Safety 1994; CSC 1997	Safety standards; civil liability for nuclear damage	Party; CSC ratified 2016; SHANTI Act 2025
Bilateral investment treaties	Protect foreign energy investors; arbitration	Most first-generation BITs terminated after 2016; Model BIT

Custom, general principles, and soft law: - **Custom:** Permanent sovereignty over natural resources (UNGA Res. 1803 (XVII), 1962; *Texaco v. Libya*, 1977); the no-harm principle (*Trail Smelter* 1941; *Pulp Mills*, ICJ 2010); duty of transboundary impact assessment. - **General principles:** Good faith, *pacta sunt servanda*, polluter pays. - **Internationalised domestic law:** The UK/Norway unbundling model travelled via the EU to India’s Electricity Act 2003; the Indonesian PSC travelled to over fifty countries. - **Soft law:** IEA and IAEA guidance, ISO 50001, Equator Principles, the WEC trilemma framework, SDG 7. Not enforceable in court, but incorporated into contracts, licences, and loan covenants.

Domestic sources in India:

Source	Examples
Constitution	Seventh Schedule; Art. 21 (clean environment; climate, per <i>M.K. Ranjitsinh</i> 2024); Art. 39(b); Art. 48A; Art. 51A(g)
Statutes	Electricity Act 2003; Energy Conservation Act 2001 (amended 2022); SHANTI Act 2025; MMDR Act 1957 (amended 2023, 2025); ORD Act 1948; PNGRB Act 2006; Essential Commodities Act 1955
Delegated legislation	CERC tariff regulations; Indian Electricity Grid Code; Electricity (Rights of Consumers) Rules 2020; Carbon Credit Trading Scheme 2023
Policy under statute	National Electricity Policy 2005 and Tariff Policy 2016 under s. 3, Electricity Act
Precedent and administration	Supreme Court, High Courts, APTEL, NGT; ministry guidelines and scheme documents

1.8 Leading Indian Precedents (Exam-Critical)

Case	Principle
<i>PTC India v. CERC</i> (2010) 4 SCC 603	CERC regulations under s. 178 are legislative; the Commission cannot depart from them in adjudication
<i>RNRL v. RIL</i> (2010) 7 SCC 1	Natural gas is held by the Government in trust for the people; private arrangements cannot override the PSC and gas policy
<i>G. Sundarajan v. UoI</i> (2013) 6 SCC 620	Kudankulam upheld; energy needs balanced against safety via the precautionary principle
<i>Hindustan Zinc v. RERC</i> (2015) 12 SCC 611	RPO may be imposed on captive generators
<i>Energy Watchdog v. CERC</i> (2017) 14 SCC 80	Foreign coal price rise is not force majeure; relief only under “change in law”
<i>M.K. Ranjitsinh v. UoI</i> 2024 INSC 280	Right against adverse effects of climate change (Arts 14, 21)

How to use these in an exam: Each case illustrates a different aspect of energy law’s **interdisciplinary and multilevel character**: - *PTC India*: the relationship between legislative and adjudicatory functions of regulators. - *RNRL*: the tension between private contract and public trust over natural resources. - *G. Sundarajan*: the precautionary principle as a tool for balancing energy security against safety. - *Hindustan Zinc*: the constitutional validity of RPO as a regulatory instrument. - *Energy Watchdog*: the limits of contractual risk allocation in a changing regulatory environment. - *M.K. Ranjitsinh*: the emergence of climate rights as a constitutional principle.

PART B – The Energy Trilemma

2.1 Origins of the Trilemma

A **trilemma** is three desirable objectives that cannot all be fully achieved at once; pursuing two compromises the third (cf. the Mundell-Fleming “impossible trinity” in economics).

Developed by the **World Energy Council** (founded 1923; a UN-accredited body spanning about 100 countries). The three objectives every energy system must pursue: 1. **Energy security** 2. **Energy equity** (access and affordability) 3. **Environmental sustainability**

The **World Energy Trilemma Index** has been published annually since 2010; the 2024 edition ranked 126 countries.

The 2024 edition’s central framing: “Evolving with Resilience and Justice.” The three dimensions must be read against overlapping crises: COVID-19 (2020), the Ukraine price shock (2022), climate-driven grid failures, and for India the West Asia crisis of 2026. The WEC explicitly recognises that the dimensions are **evolving**: - **Security** now includes the reliability of weather-dependent renewables and the availability of critical minerals. - **Equity** now includes justice in the transition itself, not only access. - **Sustainability** now extends beyond carbon to water, land, biodiversity, and waste.

This is consistent with the Heffron-Talus “fifth stage” insight: the trilemma is not a static framework but one that expands as new concerns (waste, critical minerals, cyber-resilience) emerge.

2.2 Pillar One: Energy Security

IEA definition: “The uninterrupted availability of energy sources at an affordable price.”

APERC “four A’s” (2007): Availability (do resources exist?), Accessibility (can they be reached?), Affordability (at a bearable price?), Acceptability (socially and environmentally tolerable?).

Traditional	Contemporary
Reliability: keeping the lights on; adequate generation and grid	Reliability of renewables: storage, demand response, interconnection
Physical availability: domestic resources or secure imports	Critical minerals (lithium, cobalt, nickel, rare earths, graphite) concentrated in few countries
Diversification of fuels, suppliers, routes; strategic reserves	Supply-chain security for modules, batteries, electrolyzers
Protection against physical attack and disaster	Cyber-resilience of digital grids; climate resilience

Illustrations:

- **2012 Indian blackout** (30–31 July): Northern, Eastern, and North-Eastern grids collapsed, affecting an estimated 620 million people in over twenty States — the largest blackout in history. The enquiry found States overdrawing in breach of the Indian Electricity Grid Code, inter-regional lines under maintenance, and protection systems that failed to isolate the fault. The response: stronger powers for load despatch centres and tighter frequency bands. **Reliability is a legal as much as an engineering problem.**
- **Texas, February 2021:** Winter Storm Uri froze un-weatherised gas wells, pipelines, and power plants. Over four million Texans lost power for days; more than two hundred died. ERCOT’s grid is deliberately isolated from the rest of the United States to avoid federal regulation, so Texas could not import power. The legislature then mandated weatherisation. **The security cost of a regulatory choice made for other reasons.**

- **Europe 2022:** Before 2022 Russia supplied roughly 40% of the EU's gas. When Nord Stream flows were cut, wholesale gas prices rose more than tenfold at the peak. Germany, Austria, and the Netherlands restarted or extended coal plants; the EU adopted REPowerEU (May 2022); Germany built floating LNG terminals in under a year. **Security was preserved at a cost to sustainability (more coal) and equity (record bills, mitigated by price caps and windfall taxes). The textbook trilemma trade-off.**

India's energy security profile (from PPT):

Indicator	Position
Crude oil import dependence	88.7% (FY 2025–26)
Natural gas import dependence	≈ 50%, mostly LNG
Crude supplier countries	≈ 40, up from 27
Crude via Hormuz	≈ 30%, down from about 70%
Strategic Petroleum Reserve, Phase I	5.33 million tonnes (Visakhapatnam, Mangaluru, Padur) — about 9.5 days of net imports
SPR Phase II (approved)	6.5 million tonnes (Chandikhol, Odisha; Padur) — 11.83 million tonnes total
Peak electricity demand	Record 270.8 GW met on 21 May 2026
Non-fossil installed capacity	Crossed 50% in mid-2025, five years ahead of the 2030 NDC target

Critical minerals: The DRC produces about 70% of the world's cobalt; China refines most of the world's lithium, cobalt, and graphite and dominates rare-earth processing. India's response: MMDR Amendment Act 2023 listed 24 critical and strategic minerals for Union auction; the National Critical Mineral Mission (Cabinet, January 2025) with ₹16,300 crore from the Government plus about ₹18,000 crore expected from PSUs.

Legal tools of energy security:

Instrument	Legal basis	Function
Strategic Petroleum Reserves	ISPRL (SPV under MoPNG)	Buffer against disruption
Essential Commodities Act 1955	Petroleum products as essential commodities	Control of supply in emergencies (invoked March 2026)
Electricity Act 2003, s. 11	Directions in “extraordinary circumstances”	Compelled imported-coal plants to run in 2022 and 2024
Indian Electricity Grid Code	CERC regulations under s. 79	Frequency discipline; scheduling; deviation settlement
PNGRB Act 2006, ss. 11, 20	Pipeline and city-gas authorisation	Common-carrier access
Energy Conservation Act 2001	Standards and labelling; PAT	Reduces demand and import need

2.3 Pillar Two: Energy Equity

Can all members of society access modern energy services, and can they afford to use them?

Three elements: accessibility (a physical connection), affordability (adequate use without sacrificing other necessities), universality (the last household: poor, rural, tribal, remote).

SDG 7: “Ensure access to affordable, reliable, sustainable and modern energy for all” by 2030.

Traditional	Contemporary
Universal physical access to electricity and clean cooking	Energy justice: fairness in decision-making and outcomes
Affordability of tariffs and fuel prices	Fair sharing of the benefits of the clean transition (jobs, clean air, savings)
Cross-subsidies and lifeline tariffs for the vulnerable	Just transition for fossil-fuel-dependent communities and workers
Quality and reliability of supply for all	Gender: the biomass-cooking burden on women and girls

Global picture (Tracking SDG 7, 2025): $\approx$ 655 million people without electricity; $\approx$ 2.1 billion without clean cooking fuels and technologies; $\approx$ 6.7 million deaths a year from ambient and household air pollution (WHO). Household air pollution from wood, dung, and crop residue is a leading cause of respiratory disease, borne disproportionately by women and children.

Accessibility in India: - DDUGJY (2014): last un-electrified villages connected by April 2018. - Saubhagya (2017): 2.86 crore households connected by March 2019. - PMUY (2016): over 10 crore LPG connections to poor households; ₹300 subsidy per 14.2 kg cylinder for up to nine refills in 2025–26. - **But access is not use:** NFHS-5 indicates roughly four in ten households (over half in rural India) still cook mainly with solid fuels (“fuel stacking”), because refills are unaffordable or distant.

Affordability: “energy burden”. Energy burden = share of household income spent on energy. US literature (Fisher, Sheehan and Colton) treats over 6% as high and over 10% as severe; India has no official threshold but the concept applies. Energy is a necessity with few substitutes, so the burden falls hardest on the poor; rising temperatures are turning cooling from a luxury into a survival need.

Legal tools of energy equity:

Instrument	Legal basis	Function
Universal service obligation	Electricity Act ss. 43 (duty to supply) and 6 (rural electrification)	Legal right to a connection
Cross-subsidies	ss. 61(g), 62(3); Tariff Policy 2016 ($\pm 20\%$ of average cost)	Domestic and agricultural tariffs below cost
Direct subsidy	s. 65: State must pay the licensee in advance	Free or subsidised power for farmers and the poor
Consumer rights	Electricity (Rights of Consumers) Rules 2020	Time-bound connections; reliability; prosumers
Access schemes	Saubhagya; PMUY; PM Surya Ghar	Physical access; bill relief

Cross-subsidisation: how it works: - Industrial and commercial users pay **above cost**. - Domestic and agricultural users pay **below cost**. - The DISCOM is the intermediary.

Born of equity: affordable power for farmers and the poor. **Cost:** industrial tariffs of ₹7–10 per unit push large users to captive plants and open access; the $\pm 20\%$ Tariff Policy band is breached; DISCOM borrowings $\approx$ ₹7.26 lakh crore (March 2025, PFC).

Draft Electricity (Amendment) Bill, 2025 (released by Ministry of Power for public comment on 9 October 2025): Proposals include elimination of cross-subsidies for manufacturing, railways, and metro railways within five years; tariffs that “reflect the cost of supply”; hidden cross-subsidies replaced by transparent, budgeted subsidies under s. 65; continued protection for farmers and low-income households. **Status:** still a draft at the time of writing; not enacted.

From equity to energy justice: Contemporary scholarship deepens “equity” into “energy justice” with three tenets (developed in Part C).

2.4 Pillar Three: Sustainability

Energy systems must operate within the planet’s ecological limits; the energy sector produces roughly three-quarters of global greenhouse-gas emissions.

Traditional	Contemporary
Reducing greenhouse-gas emissions	Planetary health: biodiversity, land, water
Controlling local air and water pollution	Circular economy: recycling panels, batteries, blades; decommissioning
Protecting ecosystems from extraction and infrastructure	Water–food–energy nexus: cooling water; biofuels versus food
Impact assessment before projects	Decarbonisation that itself respects ecological limits (siting)

India’s climate commitments:

Commitment	Content	Status
Updated NDC (2022)	45% cut in emissions intensity of GDP by 2030 (vs 2005); 50% non-fossil installed capacity by 2030	Capacity target achieved mid-2025
NDC for 2031–2035 (2026)	47% intensity reduction; 60% non-fossil capacity by 2035	Approved by the Union Cabinet
Net zero	Net-zero emissions by 2070	Announced at COP26 (2021)
Domestic carbon market	Energy Conservation (Amendment) Act 2022; Carbon Credit Trading Scheme 2023 (BEE)	Being implemented in phases

Local pollution: coal and air quality. Coal still supplies roughly 70% of India’s electricity generation; thermal plants emit particulates, sulphur dioxide, and oxides of nitrogen. MoEFCC notified stringent thermal emission norms in December 2015; deadlines for flue-gas desulphurisation were extended repeatedly and relaxed for most plants in 2025: **the cost of compliance (equity, security) against sustainability.** The Supreme Court’s continuing mandamus in *M.C. Mehta v. Union of India* on Delhi’s air led to the closure of the Badarpur plant (2018) and restrictions on diesel generators under the Graded Response Action Plan.

Legal tools of sustainability:

Instrument	Legal basis	Function
Environmental clearance	EPA 1986; EIA Notification 2006	Assessment and public hearing before thermal, hydro, mining, and large renewable projects
Renewable purchase obligation	Electricity Act s. 86(1)(e); Energy Conservation Act (2022) s. 14	Mandated demand for renewables
Carbon market	EC (Amendment) Act 2022; CCTS 2023	Price on emissions for obligated entities
Emission and efficiency standards	EP Rules 1986; 2015 thermal norms; Energy Conservation Act	Limits on pollutants; lower demand
Judicial oversight	NGT Act 2010; Articles 32 and 226	<i>M.C. Mehta; M.K. Ranjitsinh</i>

Sustainability versus sustainability: Decarbonisation can collide with other environmental values. The Great Indian Bustard litigation pitted solar and wind expansion against a critically endangered bird. Himalayan hydro (Subansiri, Teesta) pits low-carbon power against seismic risk, sediment, and displacement. Lithium extraction in Chile’s Salar de Atacama pits global electrification against the water rights of indigenous Atacameño communities. **Sustainability is a bundle of values that must itself be balanced.**

2.5 The World Energy Trilemma Index

Published by the World Energy Council with Oliver Wyman since 2010; 2024 was the fifteenth edition, covering 126 countries.

Each dimension scored from a basket of indicators (IEA, World Bank, IMF data); a country-context adjustment reflects governance and macroeconomic stability. Outputs: overall rank and score; a letter grade (A–D) per dimension; a “balance triangle” whose shape shows the neglected dimension; time series from 2000.

Top ten, 2024: Denmark (1), Sweden (2), Finland (3), Switzerland (4), Canada (5), Austria (6), France (7), Germany (8), Estonia (9), United Kingdom (10).

Leaders share: high low-carbon shares (wind in Denmark; hydro and nuclear in Sweden and Finland; hydro in Switzerland and Canada), strong interconnection, and high incomes that make energy affordable.

India: marked improvement on equity (access schemes), weaker on security (import dependence) and sustainability (coal), placing it in the lower-middle of the ranking.

National Energy Trilemma Index (NETI) 2024: Weights: security 25%, equity 25%, sustainability 25%, state context 25%. Top states: Kerala (65.08), Gujarat (62.71), Karnataka (61.33), Tamil Nadu (58.70), Himachal Pradesh (57.95), Andhra Pradesh (57.60), Haryana (57.12), Uttarakhand (57.10), Maharashtra (57.07), Goa (57.06).

What NETI shows: Dimension leaders: Gujarat (security and equity), Kerala (environmental sustainability), Karnataka (state context). The gap between top and bottom is wide: national averages conceal as much as they reveal. This is why the Electricity Act’s federal design matters for the trilemma: distribution, subsidies, and RPO enforcement are State functions.

Other measures: Energy Transition Index (WEF); Tracking SDG 7 (IEA, IRENA, UNSD, World Bank, WHO); World Energy Outlook (IEA); Climate Change Performance Index (Germanwatch); State Energy Efficiency Index (BEE and AEEE).

Limits of measurement: Weights are subjective; data lag by two to three years. Indices reward wealth: rich countries find affordability easy. A country is treated as one unit, hiding regional inequality (hence NETI). Resilience is invisible until a crisis reveals it. **Use an index as a diagnostic map of where a legal system's attention lies, not as a verdict.**

2.6 How Laws Balance the Trilemma

The long title of the Electricity Act 2003 pursues all three goals at once: “promoting competition ... protecting interest of consumers and supply of electricity to all areas, rationalisation of electricity tariff, ensuring transparent policies regarding subsidies, promotion of efficient and environmentally benign policies.”

- Competition and rational tariffs → security and efficiency.
- Supply to all areas and transparent subsidies → equity.
- Environmentally benign policies → sustainability.

Mechanisms: serve and strain:

Mechanism	Serves	Strains
Renewable purchase obligation	Sustainability	Equity (tariffs); security (intermittency)
Cross-subsidy	Equity	Security (DISCOM finances); sustainability (waste)
Strategic petroleum reserves	Security	Cost to the exchequer
Environmental clearance	Sustainability; procedural justice	Security (delay)
Tariff regulation (ss. 61–63)	Equity and security	Sustainability if carbon under-priced
Feed-in tariffs / competitive bidding	Sustainability; investment	Equity if high cost passed through
Grid code; must-run status	Security	Sustainability if coal preferred
Emergency directions (s. 11; ECA 1955)	Security	Sustainability; price distortion

Why the balance so often fails:

Reason	Explanation	Illustration
Crisis versus long-term goals	Security and affordability win in a shock, locking in fossil assets	Europe's 2022 coal restarts; India's 2026 kerosene return
Jurisdictional fragmentation	Ministries and regulators pursue different mandates	Coal (output) v. MoEFCC (emissions) v. Power (tariffs) v. MNRE
Political economy	Cheap power wins elections; tariff hikes lose them	Free agricultural power; deferred tariff revisions
Path dependency	Sunk coal plants, mines, and jobs create inertia	Coal districts of Jharkhand, Chhattisgarh, Odisha
Distributional conflict	Costs fall on some, benefits accrue elsewhere	Coal-royalty States v. renewable-rich Rajasthan and Gujarat
Resource constraints	Cannot invest in all three at once	SPR far below 90 days; FGD deferred on cost

Germany's Energiewende: After Fukushima (2011) Germany legislated a nuclear exit, closing its last three reactors in April 2023 while expanding wind and solar. When Russian gas was cut in 2022, the loss of firm low-carbon capacity meant lignite generation rose and prices hit records. **A legally binding sustainability commitment produced unintended security and sustainability consequences when circumstances changed.**

South Africa's load shedding: Eskom, burdened by debt, ageing coal plants, and governance failures, imposed rolling blackouts that peaked in 2023 at more than 300 days of outages. The wealthy installed solar, batteries, and generators; townships bore the cuts: "energy apartheid." **A security failure became an equity failure, while extending coal-plant lives undermined sustainability.**

Cross-subsidy: the trilemma in miniature: - **Energy equity (+):** Affordable power for farmers and low-income households; a social necessity, not merely an inefficiency. - **Energy security (-):** DISCOM borrowings ≈ ₹7.26 lakh crore (March 2025); no money for the network or adequate peak power; large consumers leave the grid. - **Environmental sustainability (-):** Free power encourages waste and groundwater over-pumping; weak DISCOMs slow to integrate renewables.

The way forward: a phased, targeted approach: - Phase out gradually: abrupt withdrawal causes hardship and backlash (the Draft Bill's five-year horizon). - Target precisely: subsidise the poor and marginal farmers, not every domestic and agricultural consumer. - Make subsidies transparent: s. 65 advance payment; direct benefit transfer (LPG under PAHAL; Punjab's agricultural power pilot). - Reinvest recovered revenue in grid modernisation, storage, and renewable integration. - Build consensus with States, farmers, industry, and DISCOM staff: electricity is a Concurrent subject and distribution a State function.

PART C — Energy Justice: The Human Dimension

3.1 What Energy Justice Adds

Developed by Sovacool, Heffron, McCauley, Jenkins, and others; deepens the equity pillar of the trilemma. Energy justice asks: **who benefits and who bears the costs; who decides and who is excluded; whose identities are recognised, at every stage from mining to waste.**

The energy justice framework maps onto legal doctrine: Articles 14, 38, 39 (distribution); natural justice, EIA public hearings, Forest Rights Act consent (procedure); Fifth and Sixth Schedules, PESA 1996, UNDRIP (recognition).

Heffron and Talus emphasise that energy justice is not merely a normative add-on but a **theoretical framework** that can guide legal analysis. It allows scholars and practitioners to evaluate whether a given energy law or policy is **just** – not just whether it is efficient or legally valid.

3.2 The Three Tenets

Tenet	Central question	Indian legal anchors
Distributive	Are benefits and burdens fairly shared?	Arts 14, 38, 39(b); District Mineral Foundation (s. 9B MMDR); rehabilitation
Procedural	Are decisions transparent, participatory, with remedies?	EIA public hearing; gram sabha consent (FRA 2006, PESA); s. 64 Electricity Act; NGT
Recognition	Are the rights and cultures of all groups acknowledged?	Fifth and Sixth Schedules; <i>Samatha</i> (1997); <i>Niyamgiri</i> (2013)

Distributive justice: lithium in the Atacama. Chile holds the world’s largest lithium reserves. SQM and Albemarle pump brine from beneath the Salar de Atacama and evaporate it. The world gets cheaper EV batteries; companies and the State get revenue. Indigenous Atacameño communities who depend on scarce water bear falling water tables and degraded wetlands. Courts and the regulator have imposed sanctions and consultation; the 2023 National Lithium Strategy proposes State participation and benefit sharing.

Procedural justice: Kudankulam. Fishing communities near the Kudankulam plant in Tamil Nadu protested after Fukushima, arguing they had not been consulted on safety, evacuation, and spent fuel. In *G. Sundarajan v. Union of India* (2013) 6 SCC 620, the Supreme Court upheld commissioning, balancing energy needs against safety through the precautionary principle, and directed safety measures, spent-fuel storage, and local involvement. **The case shows the limits of judicial review as a forum for participation.**

Recognition justice. In the Atacama the deeper injustice is that the Atacameño view of water and land as sacred was never given weight in decisions framed in tonnes and emissions. India’s coal districts (Jharkhand, Odisha, Chhattisgarh, West Bengal) hold large populations of informal workers (contract miners, loaders, transporters) absent from payrolls; just-transition plans counting only formal employees leave them behind. Jharkhand set up India’s first Sustainable Just Transition Task Force in 2022.

3.3 The Cycle of Energy Injustice

ENERGY POVERTY (no access to clean energy)

↓ forced onto costly fuels

ENERGY BURDEN (high share of income)

↓ under-use, fuel stacking

ENERGY INSECURITY (unreliable supply)

↓ democracy breaks the cycle

ENERGY DEMOCRACY (community control)

The four components defined:

Component	Definition	Indian illustration
Energy poverty	No access to modern, clean, safe energy for basic needs	About four in ten households still cook mainly with biomass
Energy burden	High share of income spent on energy	Nagpur, April 2026: ₹1,960 bill ≈ a tenth of income
Energy insecurity	Cannot rely on nominal supply: outages, voltage, disconnection	Non-solar-hour shortfalls in the May 2026 heatwave (270.8 GW peak)
Energy democracy	Community control over production, distribution, use	Solar cooperatives; panchayat storage; P2P trading pilots

Energy democracy: SPICE, Dhundi (Gujarat). In 2016 six farmers in Dhundi village, Kheda district, formed the world’s first Solar Pump Irrigators’ Cooperative Enterprise with the IWMI-Tata programme. Solar pumps replaced diesel; surplus power is sold to the State DISCOM (MGVCL) under a 25-year PPA at about ₹4.63 per unit. **Solar becomes a “remunerative crop”; over-pumping of groundwater is discouraged because pumping now has an opportunity cost; farmers, not a distant utility, control the asset.**

Three more examples: - **Modhera** (Mehsana, Gujarat) became India’s first round-the-clock solar village in 2022: about 1,300 rooftop systems, a 6 MW plant, and a 15 MWh battery. MNRE has earmarked ₹800 crore for one Model Solar Village per district (₹1 crore each). - **Perinjalam panchayat** (Thrissur, Kerala) has put rooftop solar on about a thousand homes since 2017 via a consumers’ committee and cooperative-bank loans, and is building India’s first panchayat-level battery storage with ₹2 crore of State support. - **UPERC peer-to-peer trading pilot** (23 February 2026): India’s first interstate P2P renewable trading pilot (PVVNL area to Delhi DISCOM consumers) on a blockchain platform under the India Energy Stack, wheeling charge ₹1.01 per unit; extended to March 2027.

Policies for the four components:

Component	Policy	Key features
Energy burden	PM Surya Ghar (2024); AP weavers; Maharashtra; Bihar	Rooftop solar and free units for low-income and occupational groups
Energy insecurity	Emergency measures, March 2026	LPG directions; ECA for gas; PDS kerosene in 21 States/UTs; diesel cap
Energy poverty	PMUY (2016); Saubhagya (2017)	Free LPG connection and first cylinder; ₹300 refill subsidy; 2.86 crore households electrified
Energy democracy	Model Solar Village; UPERC P2P pilot; cooperatives	Community ownership of generation and storage

How the components interact: the PMUY cycle: - **Energy poverty:** millions of rural households cooked with biomass. - **PMUY** provided free connections, removing the access barrier. - **Energy burden:** even a subsidised refill is a large outlay for a household earning a few thousand rupees a month. - **Energy insecurity:** households “stack” fuels, reverting to biomass when cash is short, so the health benefit is only partly realised. - **Energy democracy:** community-scale solutions (biogas

cooperatives, collective LPG procurement, electric cooking on rooftop solar) address what connection alone cannot.

3.4 Two Logics (Mohan and Topp, 2018)

The PPT presents this as a binary, but the source article makes clear that these are **contested narratives** in Indian energy policy, not mutually exclusive positions.

Feature	“Energy for growth”	“Energy for all”
Conception of energy	A productive input for industry and GDP	A basic human need and condition of dignity
Central question	How do we keep energy from constraining growth?	How do we ensure no one is left behind?
Governance style	Centralised, technocratic, top-down	Participatory, decentralised, rights-based
Investment focus	Ultra-mega plants, national grid, pipelines	Last-mile connections, decentralised renewables
View of cross-subsidy	A distortion that drives large consumers off-grid	A necessary social intervention

Source context: Mohan and Topp (2018) identify two dominant narratives in Indian energy policy debates: “energy for development” that privileges energy as critical to economic growth and long-term strategic security, and “energy for all” that prioritises the role of energy for basic development and ending poverty. While these narratives find common ground on certain issues such as the role of coal, they clash in the socio-technical imaginaries they represent about India’s energy future.

Why this matters for the exam: The trilemma (security, equity, sustainability) is often presented as a technical balancing exercise. Mohan and Topp show that the “balance” is actually a **political contest** between competing narratives about what energy is for. Energy law is the terrain on which this contest is fought — through RPO regulations, tariff orders, environmental clearances, and just-transition plans.

Appendix: Key Data for Quick Revision

Evolution stages (Heffron & Talus): 1. Safety (1820s) — mining accidents 2. Security (1910s) — preventing war (ECSC, Euratom) 3. Economics (1960s) — liberalisation, competition 4. Infrastructure (2000s) — grids, pipelines 5. Justice (emerging) — fairness, waste

Trilemma dimensions: - Security: reliability, availability, resilience - Equity: access, affordability, universality - Sustainability: decarbonisation, pollution, limits

Energy justice tenets: Distributive · Procedural · Recognition

Two logics: “Energy for growth” vs. “Energy for all”

Key Indian cases: *PTC India · RNRL · G. Sundarajan · Hindustan Zinc · Energy Watchdog · M.K. Ranjitsinh*

Key international instruments: UNCLOS 1982 · Energy Charter Treaty 1994 (India not a party) · Paris Agreement 2015 · Convention on Nuclear Safety 1994